

## Implementation:

To implement the Iceberg tables, we need to create **External Volumes**:

External Volumes hold the File data and table metadata, all in customer-supplied storage.

### □ Create External Volume:

- Before creating External Volume, Create Bucket and new Role on your AWS account
- Grant S3 full access policy to your new Role

Click the **Create role** button.

The screenshot shows the AWS IAM console 'Create role' page. The 'Select type of trusted entity' section has four options: 'AWS service' (EC2, Lambda and others), 'Another AWS account' (Belonging to you or 3rd party), 'Web identity' (Cognito or any OpenID provider), and 'SAML'. The 'Another AWS account' option is selected. Below this, a note states: 'Allows entities in other accounts to perform actions in this account. [Learn more](#)'. The 'Specify accounts that can use this role' section has a text input for 'Account ID\*' with the value '123456789000'. The 'Options' section has a checkbox for 'Require external ID' which is checked. Below this, a text box for 'External ID' contains the value '0000'. A help text box explains: 'You can increase the security of your role by requiring an optional external ID. This prevents "confused deputy" attacks. This is recommended if you do not have administrative access to the account that can assume this role. The external ID can be any characters that you choose. To assume this role, users must be required to provide this exact external ID. [Learn more](#)'.

## Create External Volume pointing to the S3 bucket

create external volume iceberg\_int

storage\_locations =

(

(

name = 'my-s3-us-east-2'

storage\_provider = 'S3'

storage\_base\_url = 's3://icebergconfigbucket/'

storage\_aws\_role\_arn = 'arn:aws:iam::913267004595:role/icebergconfigrole'

)

);

### ❑ DESC EXTERNAL VOLUME iceberg\_int and copy the below highlighted parameters

```
{"NAME":"my-s3-us-east-2","STORAGE_PROVIDER":"S3",
```

```
"STORAGE_BASE_URL":"s3://icebergpracticebucket1/","STORAGE_ALLOWED_LOCATIONS":["s3://icebergpracticebucket1/*"],
```

```
"STORAGE_REGION":"us-east-2","STORAGE_AWS_ROLE_ARN":"arn:aws:iam::913267004595:role/icebergrole",
```

```
"STORAGE_AWS_IAM_USER_ARN":"arn:aws:iam::315033924470:user/i4v10000-s",
```

```
"STORAGE_AWS_EXTERNAL_ID":"QX73654_SFCTestRole=2_f2MhffRI7tL5Dlrr1HpqmsLmcwc="}
```

### ❑ Granting the IAM User Permissions to Access Bucket Objects

- Click the Edit trust relationship button of your new role.

- Modify the policy document with the DESC EXTERNAL VOLUME output values you recorded in previous step
- Click the Update Trust Policy button. The changes are saved

## □ Create Iceberg table: Two ways to create iceberg table:

### Approach1:

```
create or replace iceberg table customer_detail (
  CUST_NUM varchar,
  CUST_STAT varchar ,
  CUST_BAL number(10,0),
  INV_NO varchar ,
  INV_AMT number(10,2),
  CRID varchar ,
  SSN varchar,
  phone number(10,0),
  Email varchar
)
CATALOG = 'SNOWFLAKE'
external_volume='iceberg_int'
BASE_LOCATION = 'CUSTOMER_INFO';
```

### **create or replace storage integration s3\_int**

**type = external\_stage**

**storage\_provider = s3**

**enabled = true**

**storage\_aws\_role\_arn = 'arn:aws:iam::913267004595:role/testsnowflake'**

**storage\_allowed\_locations = ('s3://srcbucketsachin/');**

**desc integration s3\_int**

**create or replace file format demo\_db.public.csv\_format**

```
type = csv
skip_header = 1
null_if = ('NULL', 'null')
empty_field_as_null = true;
```

**create or replace stage demo\_db.public.ext\_csv\_stage**

```
URL = 's3://srcbucketsachin/POC'
STORAGE_INTEGRATION = s3_int
file_format = demo_db.public.csv_format;
```

**list @demo\_db.public.ext\_csv\_stage**

**copy into prod\_tbl**

**from @demo\_db.public.ext\_csv\_stage/Customer\_Inv.csv**

**on\_error = CONTINUE;**

#### **Approach2:**

```
create or replace iceberg table customer_detail
with EXTERNAL_VOLUME = 'iceberg_int'
as select CUST_NUM, CUST_STAT, CUST_BAL, INV_NO, INV_AMT,
CRID, SSN, PHONE, EMAIL from CUSTOMER;
```